Master of Business Analytics

Prescriptive Analytics & Optimization

MBAN 5540

Class 05: Integer Programming

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Integer Programming (IP)

- In LP, we assume that fractional values are acceptable.
- In real-world problems, sometimes only Integer values are acceptable.
- There are 2 types of IP:
  - General IP: variables can take any non-negative integer values such as 0,1,2,… shown as $\mathbb{Z}^+$
  - Binary IP: variables can only take 0 and 1

Integer Programming

- Pure Integer Programming (IP) Problems: when the problem only contains integer variables

- Mixed-Integer Programming (MIP) Problems: when there are both integer and non-integer variables.
  - Most of the real-world problems are MIP.

Does Rounding Work?

- One may suggest solving the LP version of the problem, and then rounding the fractional values
  - Consider the following problem

$$
\begin{aligned}
Max \quad Z = x + 0.64y \\
50x + \quad 31y \leq 250 \\
-3x + \quad 2y \leq 4 \\
x, y \in \mathbb{Z}^+
\end{aligned}
$$

  - The LP solution : $x = 1.94 \quad y = 4.92 \Longrightarrow \quad Z = 5.09$
  - The IP Solution : $x = 5 \quad y = 0, \quad \Longrightarrow \quad Z = 5$

  - It may work, but not always !!
  - Rounding might make the solution infeasible!!

Solving IP

- Now, consider the following problem

$$
\begin{aligned}
Max\ Z = 600x + 700y \\
2x + 3y \leq 12 \\
6x + 5y \leq 30 \\
x, y \in \mathbb{Z}^+
\end{aligned}
$$

- If we change $x, y \in \mathbb{Z}^+$ to $x, y \geq 0$, we call the new problem the <u>linear relaxation</u> of the original problem.

IP Vs. LP

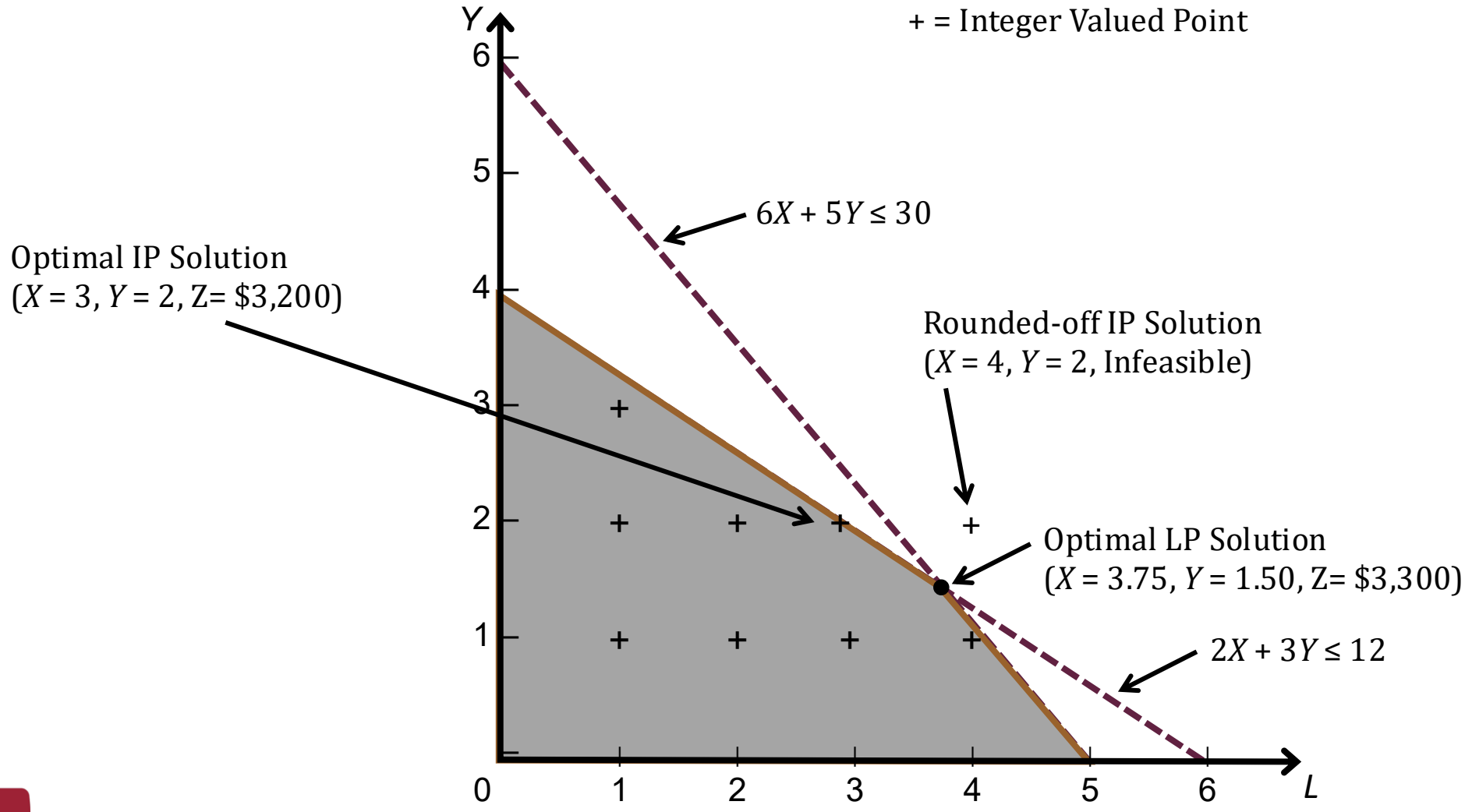
- The LP optimal solution is:
- $x = 3.75, \quad y = 1.5 \quad \Longrightarrow \quad Z: 3{,}300$

- If we round to $x = 4, \quad y = 2$, the solution is not feasible.
- If we round to $x = 3, \quad y = 1$, the solution is 1,800.
- If we round to $x = 4, \quad y = 1$, the solution is 3,100.

- The IP optimal solution is:
- $x = 3, \quad y = 2 \qquad \Longrightarrow \quad Z = 3{,}200$

IP solution CANNOT be better than LP solution.

Graphical Solution



Important Remarks

1. The IP optimal solution is not on a corner point. Therefore, the Simplex method of Solver does not work.

2. Solver uses a method called “Branch-and-Bound” or “B&B”. It searches the feasible region to find the optimal integer solution.

3. In changing an LP problem to IP problem, more restrictions will be imposed on the problem. Therefore, the feasible region will be smaller, and the IP solution cannot be better. The IP solution is either worse than LP solution or it is the same solution.

Binary Variables

- Can only take 0 and 1 values
- Very helpful in modelling
  - Yes/No decisions
  - Selection decisions
  - Covering decisions

Some Binary Applications

- Investment Analysis
  - Shall we make a certain investment?
- Site Selection
  - Should a certain site be selected to locate a new facility?
- Airline Application
  - Should a certain type of airplane be assigned to a certain flight leg?
  - Should a certain sequence of flight legs be assigned to a crew?
- Production and Distribution Network
  - Should a certain plant remain open?
  - Should a certain site be selected for a new plant?
  - Should a certain distribution center be assigned to serve a certain market area?

Writing Constraints Using Binary Variables

Selecting k out of n choices ($n \geq k$)

If we define x_i as a binary variable:

- Selecting <u>exactly</u> k out of n choices:

$$x_1 + x_2 + \cdots + x_n = k$$

- Selecting <u>at least</u> k out of n choices:

$$x_1 + x_2 + \cdots + x_n \geq k$$

- Selecting <u>at most</u> k out of n choices:

$$x_1 + x_2 + \cdots + x_n \leq k$$

Managing Incompatible Selections

- Mutually exclusive: Only one of the decision variables can be selected:

$$x_1 + x_2 + \cdots + x_n = 1$$

- If one is selected, others cannot be selected:

$$x_1 + x_2 + \cdots + x_n \leq 1$$

Managing Dependencies

- If decision x_1 is selected, then x_2 must be selected as well, i.e., x_1 requirs x_2 (one way):

$$x_1 \leq x_2 \quad \text{or} \quad x_1 - x_2 \leq 0$$

  - If x_1 requirs x_2 <u>or</u> x_3:

$$x_1 \leq x_2 + x_3$$

  - If x_1 requirs x_2 <u>and</u> x_3:

$$2x_1 \leq x_2 + x_3$$

- If decision x_1 is selected, then x_2 must be selected and vice versa (two way):

$$x_1 = x_2 \quad \text{or} \quad x_1 - x_2 = 0$$

Managing Incompatible Constraints

- Sometimes, we need to select one of two constraints, with which we can get a better solution:

$$3x_1 + 2x_2 \leq 5$$
$$x_1 + 4x_2 \leq 8$$

- Introduce a binary variable y :

$$3x_1 + 2x_2 \leq 5 + My$$
$$x_1 + 4x_2 \leq 8 + M(1 - y)$$

Where M is a very large positive number.

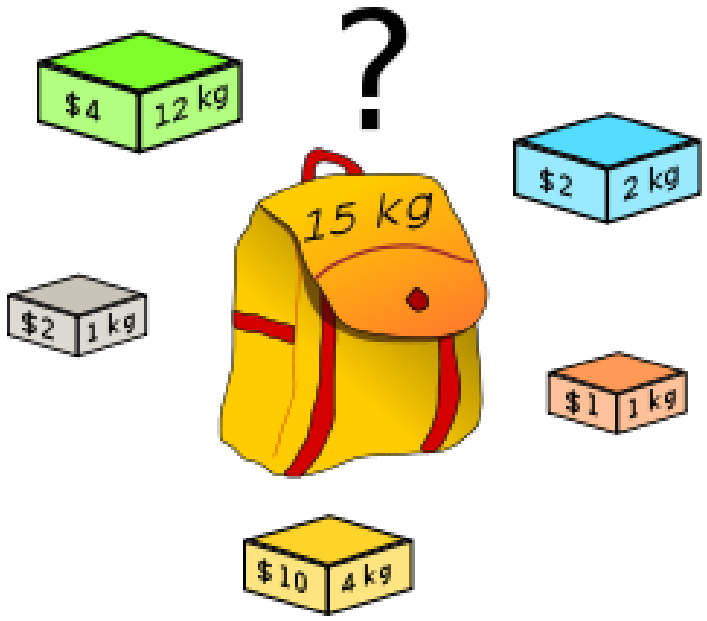
Some Applications

The Knapsack Problems

The Assignment Problems

The Knapsack Problems

- Suppose we have a set of items:
  - Each item has a value and a weight
- The goal is to maximize the total value that we put in our knapsack
- The knapsack has a given capacity.
- Many applications in resource allocation
  - E.g., SaaS (Software as a service)
  - Virtual machine allocation in cloud computing



The Knapsack Problems

Parameters

- n is the number of items
- w_i is the wight of item i
- v_i is the value of item i
- C is the capacity of knapsack

Decision Variables

- $x_i = \begin{cases} 1, & \text{if item } i \text{ is selected} \\ 0, & \text{otherwise} \end{cases}$

$$\max \quad \sum_{i=1}^{n} v_i x_i$$

$$\text{S.t.:} \quad \sum_{i=1}^{n} w_i x_i \leq C$$

$$x_i \in \{0, 1\} \qquad (i = 1, 2, \ldots, n)$$

Extension: Multiple Knapsack Problems

- It is possible that we have multiple knapsacks instead of just one:

Parameters

- n is the number of items
- m is the number of knapsacks
- w_i is the wight of item i
- v_i is the value of item i
- C_j is the capacity of knapsack j

Decision Variables

- $x_{ij} = \begin{cases} 1, & \text{if item } i \text{ is placed in knapsack } j \\ 0, & \text{otherwise} \end{cases}$

$$\max \quad \sum_{i=1}^{n} \sum_{j=1}^{m} v_i x_{ij}$$

$$\text{S.t.:} \quad \sum_{i=1}^{n} w_i x_{ij} \leq C_j \qquad (j = 1, 2, \ldots, m)$$

$$\sum_{j=1}^{m} x_{ij} \leq 1 \qquad (1 = 1, 2, \ldots, n)$$

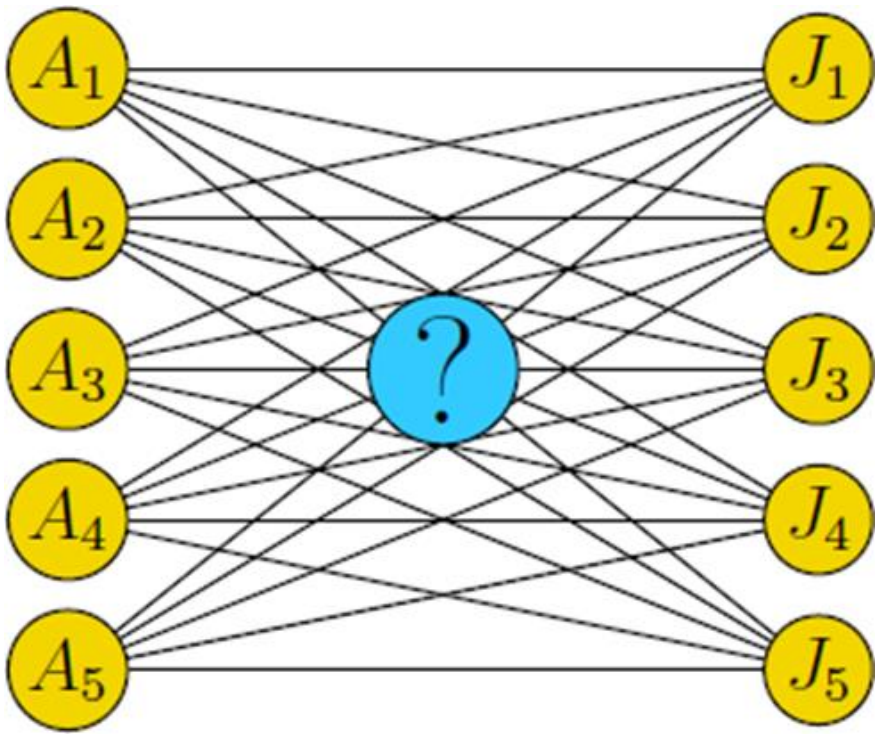
$$x_{ij} \in \{0, 1\} \qquad (i = 1, 2, \ldots, n,\ j = 1, 2, \ldots, m)$$

The Assignment Problem

- The assignment problem arises in a variety of decision-making situations:
  - assigning jobs to machines
  - agents to task
  - sales personnel to sell territories
  - contracts to bidders and so on.

- The distinguishing feature of the assignment problem is that **one** agent is assigned to **one and only one** task.

- Objective: minimizing costs, minimizing time, or maximizing profit.



Assignment Problem

Parameters

- m and n are the number of people and jobs respectively ($m \geq n$).
- c_{ij} is the cost of assigning person i to job j.

Decision Variables

- $x_{ij} = \begin{cases} 1, & \text{if person } i \text{ is assigned to job } j \\ 0, & \text{otherwise} \end{cases}$

$$\min \quad \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij}$$

$$\text{S.t.:} \quad \sum_{i=1}^{m} x_{ij} = 1 \qquad (j = 1, 2, \ldots, n)$$

$$\sum_{j=1}^{n} x_{ij} \leq 1 \qquad (i = 1, 2, \ldots, m)$$

$$x_{ij} \in \{0, 1\} \qquad (i = 1, 2, \ldots, m, \; j = 1, 2, \ldots, n)$$

Assignment Problem

- What if we can assign one person to more than one task?
- Assume, we have a maximum number (upper bound) of tasks that can be assigned to each person.

$$\sum_{j=1}^{n} x_{ij} \leq 1 \quad (i = 1, 2, \ldots, m) \longrightarrow \sum_{j=1}^{n} x_{ij} \leq UB_i \quad (i = 1, 2, \ldots, m)$$